

Cross-sectional association of nut intake with adiposity in a Mediterranean population.



BACKGROUND AND AIMS: Nut intake has been inversely related to body mass index (BMI) in prospective studies. We examined dietary determinants of adiposity in an elderly Mediterranean population with customarily high nut consumption. METHODS AND RESULTS: A cross-sectional study was conducted in 847 subjects (56% women, mean age 67 years, BMI 29.7kg/m²) at high cardiovascular risk recruited into the PREDIMED study. Food consumption was evaluated by a validated semi-quantitative questionnaire, energy expenditure in physical activity by the Minnesota Leisure Time Activity questionnaire, and anthropometric variables by standard measurements. Nut intake decreased across quintiles of both BMI and waist circumference (P-trend <0.005; both). Alcohol ingestion was inversely related to BMI (P-trend=0.020) and directly to waist (P-trend=0.011), while meat intake was directly associated with waist circumference (P-trend=0.018). In fully adjusted multivariable models, independent dietary associations of BMI were the intake of nuts inversely (P=0.002) and that of meat and meat products directly (P=0.042). For waist circumference, independent dietary associations were intake of nuts (P=0.002) and vegetables (P=0.040), both inversely, and intake of meat and meat products directly (P=0.009). From the regression coefficients, it was predicted that BMI and waist circumference decreased by 0.78kg/m² and 2.1cm, respectively, for each serving of 30g of nuts. Results were similar in men and women. CONCLUSION: Nut consumption was inversely associated with adiposity independently of other lifestyle variables. It remains to be explored whether residual confounding related to a healthier lifestyle of nut eaters might in part explain these results. Copyright © 2009 Elsevier B.V. All rights reserved.